BE II Year Examination Dec 2023

Computer Engineering (A and B)

3CERC3 Data Structure Using C

Max Marks: 60

Duration: 3 Hrs

Note: Attempt any one part from each question. Each question carries equal marks. Answer should be precise and to the point.

Q.1 a) Compare static and dynamic array also provide the implementation for both of them. Also explain array of structures with suitable example in detail.

Q.1 b) Compare linear search and binary search in terms of time complexity. Also provide implementation.

Q.2 a) Compare Merge Sort and Quick sort find their time complexity and provide the implementation.

Q.2 b) Define a linked list. Explain the implementation of operations of a linked list.

Q.3 a) Explain the concept of a circular linked list. Write the conditions how to insert, delete elements of circular linked list. Also explain the application of circular linked list in real life.

Q.3 b) Explain Stack implementation in detail with all the operations. Also provide the applications of stack.

Q.4 a) Calculate the time complexity of following:

1. $T(n)=2T(n/2)+O(n)$

2. $T(n)=1$, if $n=1$, $T(n)=T(n-2)+1$ if $n>1$.

Q.4 b) Compare Linear queue, Circular queue, Priority queue and Deque.

Q.5 a) Write a short note on Tree Traversal. Explain the concept of AVL trees and why they are used

Q.5 b) Compare BFS and DFS. Apply BFS and DFS in following figure: